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## Stacked lens, optical unit, endoscope, and method of manufacturing optical unit


### Abstract

A stacked lens includes: a first optical device; a second optical device disposed on an optical axis of the first optical device; and an adhesive layer bonding the first optical device and the second optical device and including a projection projecting outside a side surface of the first optical device and a side surface of the second optical device.



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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of PCT/JP 2021/010651 filed on Mar. 16, 2021, the entire contents of which are incorporated herein by this reference.

BACKGROUND OF THE INVENTION
1. Field of the Invention
(1) Embodiments of the present invention relate to a stacked lens including a plurality of optical devices stacked with an adhesive layer in between, an optical unit including the stacked lens, an endoscope including the optical unit, and a method of manufacturing the optical unit.
2. Description of the Related Art
(2) Optical units are used for an image pickup optical system for an image pickup unit and an illumination optical system for an illumination unit. To downsize an optical unit, a stacked lens including a plurality of optical devices stacked with an adhesive layer in between has been suggested. The stacked lens is manufactured by cutting a stacked wafer including a plurality of optical wafers bonded with an adhesive agent.
(3) International Publication No. 2018/087872 discloses an endoscope where a stacked lens, a side surface of which includes a recessed portion, is disposed in a distal end member. In inserting the stacked lens into the distal end member for fixation, a position offset is prevented by bringing the recessed portion of the side surface into contact with a projection of the distal end member.
SUMMARY OF THE INVENTION
(4) A stacked lens according to an embodiment of the present invention includes: a first optical device; a second optical device disposed on an optical axis of the first optical device; and a first adhesive layer bonding the first optical device and the second optical device and including a first projection projecting outside an outer peripheral surface of each of the first optical device and the second optical device, the outer peripheral surface being along the optical axis.
(5) An optical unit according to an embodiment of the present invention includes: a stacked lens including: a first optical device; a second optical device disposed on an optical axis of the first optical device; and a first adhesive layer bonding the first optical device and the second optical device and including a first projection projecting outside an outer peripheral surface of each of the first optical device and the second optical device, the outer peripheral surface being along the optical axis; and a first frame member in which the stacked lens is housed and that has an inner surface including a first positioning member, the first projection being in contact with the first positioning member.
(6) An endoscope according to an embodiment of the present invention includes an insertion section and an optical unit disposed in a distal end portion of the insertion section, the optical unit including: a stacked lens including: a first optical device; a second optical device disposed on an optical axis of the first optical device; and a first adhesive layer bonding the first optical device and the second optical device and including a first projection projecting outside an outer peripheral surface of each of the first optical device and the second optical device, the outer peripheral surface being along the optical axis; and a frame member in which the stacked lens is housed and that has an inner surface including a positioning member, the first projection being in contact with the positioning member.
(7) A method of manufacturing an optical unit according to an embodiment of the present invention

includes: producing a stacked wafer including a first optical wafer including a plurality of first optical devices, a second optical wafer including a plurality of second optical devices, and an adhesive layer between the first optical wafer and the second optical wafer; cutting the stacked wafer using a dicing blade to produce a stacked lens, the stacked lens including one of the first optical devices, one of the second optical devices, and a first adhesive layer including a projection projecting outside a cut surface; and housing the stacked lens in a frame member having an inner surface including a positioning member in a state where the projection is in contact with the positioning member.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of a stacked lens of a first embodiment;
(2) FIG. **2** is a side view of the stacked lens of the first embodiment;
(3) FIG. **3** is a top view of the stacked lens of the first embodiment;
(4) FIG. **4** is a cross-sectional view for explaining a method of manufacturing the stacked lens of the first embodiment;
(5) FIG. **5** is a cross-sectional view for explaining the method of manufacturing the stacked lens of the first embodiment;
(6) FIG. **6** is a perspective view of a stacked lens of Modification 1 of the first embodiment;
(7) FIG. **7** is a perspective view of a stacked lens of Modification 2 of the first embodiment;
(8) FIG. **8** is an exploded perspective view of an optical unit of a second embodiment;
(9) FIG. **9** is a perspective view of the optical unit of the second embodiment;
(10) FIG. **10** is a cross-sectional view taken along a line X-X in FIG. **9**;
(11) FIG. **11** is a flowchart of a method of manufacturing the optical unit of the second embodiment;
(12) FIG. **12** is an exploded perspective view of an optical unit of Modification 1 of the second embodiment;
(13) FIG. **13** is a perspective view of the optical unit of Modification 1 of the second embodiment;
(14) FIG. **14** is a cross-sectional view taken along a line XIV-XIV in FIG. **13**;
(15) FIG. **15** is an exploded perspective view of an optical unit of Modification 2 of the second embodiment;
(16) FIG. **16** is an appearance view of an endoscope of a third embodiment; and
(17) FIG. **17** is an exploded perspective view of the endoscope of the third embodiment.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
First Embodiment
(18) As illustrated in FIG. **1** to FIG. **3**, a stacked lens **1** of a present embodiment includes a first optical device **10**, a second optical device **20** disposed on an optical axis of the first optical device **10**, and a first adhesive layer **30** (hereinafter, referred to as an “adhesive layer **30**”) bonding the first optical device **10** and the second optical device **20**.
(19) Note that drawings of the embodiments and the like are schematic views. A relationship between a thickness and a width of each portion, a ratio between respective thicknesses of portions, and the like are different from actual ones. A relationship and a ratio between respective dimensions of some portions may be different even among the drawings. In addition, illustration and assignment of reference numbers are omitted for some components.
(20) The first optical device **10**, which is in a shape of rectangular parallelepiped, has a first principal surface **10**SA and four side surfaces **10**SS. The second optical device **20**, which is in a shape of a rectangular parallelepiped, has a second principal surface **20**SB and four side surfaces **20**SS. The adhesive layer **30** is a resin layer bonding a surface of the first optical device **10**

opposite the first principal surface **10**SA and a surface of the second optical device **20** opposite the second principal surface **20**SB. The first principal surface **10**SA and the second principal surface **120**SB are substantially in the same shape and have substantially the same surface area. That is, the four side surfaces **10**SS of the first optical device **10** are flush with the respective four side surfaces **20**SS of the second optical device **20**. In other words, the first optical device **10** and the second optical device **20** overlap as observed in a direction from the first principal surface **10**SA of the stacked lens **1**.

(21) The first optical device **10** is a hybrid lens device, where a resin lens **12** is disposed on a glass device **11**. However, in the drawings, the resin lens **12** is illustrated as a flat plate and the first optical device **10** and the second optical device **20** are illustrated as being filled with the adhesive layer **30** with no gap between. The second optical device **20** is made of a colorless and transparent flat glass but may include a colored and transparent filter or glass lens.

(22) Respective four side surfaces **30**SS of the adhesive layer **30** project from the respective four side surfaces loss of the first optical device **10** and the respective four side surfaces **20**SS of the second optical device **20**. That is, the adhesive layer **30** includes an elongated protrusion, or first projection **31** (hereinafter, referred to as a "projection **31**"), not interposed between the first optical device **10** and the second optical device **20**. As illustrated in FIG. **3**, the projection **31** surrounds the first optical device **10** (the first principal surface **10**SA) and has an outline substantially in a shape of a quadrilateral picture frame as observed in the direction from the first principal surface **10**SA of the stacked lens **1**.

(23) In other words, the adhesive layer **30** has an inner peripheral region bonding the first optical device **10** and the second optical device **20** and an outer peripheral region in a form of the projection **31** projecting outside the four side surfaces of the first optical device **10** and the four side surfaces of the second optical device **20**. That is, the projection **31** projects outside the four side surfaces **10**SS and the four side surfaces **20**SS, which are outer peripheral surfaces of the first optical device **10** and the second optical device **20** along an optical axis O.

(24) A thickness of the adhesive layer **30**, that is, a thickness of the projection **31**, is in a range from 50 μm to 100 μm. In addition, the adhesive layer **30** is made of soft resin having a smaller modulus of elasticity E than the first optical device **10** and the second optical device **20**. For example, the modulus of elasticity E of the adhesive layer **30** is preferably in a range from 0.1 MPa to 4 MPa.

(25) For example, in holding the stacked lens **1** with tweezers the soft projection **31** serves as a slip resistance; therefore, the stacked lens **1** is user-friendly.

(26) The modulus of elasticity E is measured by tensile dynamic viscoelastic measurement (DMA). After a sample is clamped to a measurement head, stress is applied to the 25-degree-C. sample from a load generation section through a probe. The stress was provided as a sinusoidal force with a predetermined frequency and set such that a strain amplitude of the sample became constant. A deformation of the sample caused by the sinusoidal force is detected by a deformation detection section. The modulus of elasticity was calculated from the stress applied to the sample and the detected strain (for example, JIS K6394).

(27) At a projection amount of the projection **31** from the side surfaces **10**SS, **20**SS of 5 μm or more, a slip resistance effect is outstanding. At a projection amount of 100 μm or less, the stacked lens **1** has a small outer size in a direction orthogonal to the optical axis O.

(28) The first optical device **10** and the second optical device **20** are optical devices configured to provide desired optical properties. For example, the optical devices may each be a hybrid lens device where resin lens devices are disposed on both sides of a glass device. That is, the adhesive layer may bond the resin lens device and the resin lens device. In addition, the adhesive layer may bond the glass device and the glass device. Note that an incoming surface of the stacked lens where light is to enter may be one of the first principal surface **10**SA and the second principal surface **20**SB.

(29) <Method of Manufacturing Stacked Lens>

(30) Description will be made on a method of manufacturing the stacked lens **1**.
(31) <Step S**10**> Production of Optical Wafers
(32) A first optical wafer **10**W illustrated in FIG. **4** is produced by disposing a resin layer **12**W, which includes a plurality of the resin lenses **12**, all over a glass wafer **11**W. The second optical wafer **20**W is a glass wafer.
(33) It is preferable that the resin layer **12**W be curable resin. When the curable resin externally receives an energy such as heat, ultraviolet light, or electron beam, a cross-linking reaction or a polymerization reaction progresses. The resin lens **12** is made of, for example, transparent ultraviolet curable silicone resin, epoxy resin, or acrylic resin. Note that “transparent” means that light absorption and diffusion of a material are small enough to be usable in a wavelength range to use.
(34) For example, the resin layer **12**W is produced by a die molding method including disposing uncured resin on the glass wafer **11**W and applying ultraviolet light to cure the resin while pressing a die having a recessed portion with a predetermined inner surface shape against the resin. Since the inner surface shape of the die is transferred to an outer surface shape of the resin lens, the die molding method enables easily producing even an aspheric lens.
(35) The resin layer **12**W including the resin lenses **12** may be produced by patterning a resin layer by a photolithography technique or a metal masking technique and then performing an appropriate heat treatment. Alternatively, the resin layer **12**W may be produced by an ink jet technique.
(36) The first optical wafer **10**W and the second optical wafer **20**W may each be transparent resin as a whole or may each be a combination of glass and resin as appropriate. That is, the first optical wafer **10**W and the second optical wafer **20**W may be composed of a plurality of layers made of the same material or different materials. In addition, the first optical wafer **10**W and the second optical wafer **20**W may each be a rectangular wafer or a round wafer.
(37) The first optical wafer **10**W and the second optical wafer **20**W only have to have respective light transmissive light path regions. For example, a light blocking aperture film may be disposed around each of light paths.
(38) <Step S**20**> Production of Stacked Wafer
(39) As illustrated in FIG. **4**, an adhesive agent **30**W is applied between the first optical wafer **10**W and the second optical wafer **20**W. By virtue of the adhesive layer **30** formed by curing the adhesive agent **30**W, the first optical wafer **10**W and the second optical wafer **20**W are fixed in a direction of the optical axis O to produce the stacked wafer **1**W. Predetermined known resin such as acrylic resin is used as the adhesive agent **30**W constituting the adhesive layer **30**. A viscosity η of the uncured adhesive agent **30**W is, for example, in a range from 400 mPa.Math.s to 3000 mPa.Math.s.
(40) Note that the adhesive layer **30** is made of transparent resin. However, the adhesive layer **30** at a position where the light path is not blocked may be made of a light blocking resin. In addition, the adhesive layer **30** made of the light blocking resin may serve as an aperture.
(41) <Step S**30**> Dicing (Cutting of Stacked Wafer)
(42) As illustrated in FIG. **5**, the stacked wafer **1**W fixed to a dicing tape **70** is diced using a dicing blade **90** and thus divided into stacked lenses **1**.
(43) According to studies having been performed so far by the present inventors, it has been found that as a result of dividing the stacked wafer **1**W into the stacked lenses **1** using the dicing blade **90** under predetermined conditions, the projection **31** of the adhesive layer **30** projects from a cut surface. A reason for the above is unknown but it is speculated that a force acting on the adhesive layer **30** in a lower direction and an external direction would be a main reason. The projection amount of the projection **31** and the number of side surfaces where the projection **31** projects could be controlled by changing a type of the adhesive agent, a thickness of the adhesive layer, the conditions for blade dicing, and the like.
(44) The stacked lens **1** where the projection **31** of the adhesive layer **30** projects after dicing is

produced by setting a type and a thickness of the adhesive layer **30**. For example, the stacked lens **1** including the projection **31** with a projection amount in a range from 5 μm to 100 μm is produced by forming the adhesive layer **30** with a thickness in a range from 50 μm to 100 μm by using acrylic resin with a modulus of elasticity E in a range from 0.1 MPa to 4 MPa.
(45) Even when the type of the adhesive layer **30** is not acrylic resin, an effect similar to the effect of the adhesive layer **30** can be achieved by, for example, adjusting the layer thickness and the modulus of elasticity.
(46) The side surface **10**SS of the first optical device **10** and the side surface **20**SS of the second optical device **20**, which are cut surfaces of the stacked lens **1** divided using the dicing blade **90**, include a striation. The striation is likely to occur in resin of the dicing tape **70** or the like but may occur in glass or the like.
(47) In other words, an optical device including a striation in a side surface can be considered as having been divided using the dicing blade **90**.
(48) Note that in a case where the stacked wafer **1**W was divided by laser dicing or etching, no projection of the adhesive layer **30** from a side surface occurred during a dividing work.
(49) A stacked lens including a projection projecting from a side surface of an optical device can also be produced by, for example, cutting the glass wafer **11**W including glass and the second optical wafer **20**W by a cutting method allowing for cutting glass but not resin, such as ultrashort pulsed-laser or hydrofluoric-acid etching, and then cutting the adhesive layer **30** with a cutting allowance smaller than a cutting allowance for glass. However, these manufacturing methods require a larger number of processes and higher manufacturing costs than the manufacturing method of the present embodiment.
(50) For example, in holding the stacked lens **1** including the projection **31** projecting from the side surface of the optical device with tweezers, the soft projection **31** serves as a slip resistance; therefore, the stacked lens **1** is user-friendly.
(51) <Modifications of First Embodiment>
(52) Since stacked lenses **1**A, **1**B of Modifications 1, 2 of the first embodiment are similar to the stacked lens **1**, the same reference sign is assigned to components with the same function and an explanation is omitted.
(53) <Modification 1 of First Embodiment>
(54) The stacked lens **1** of the first embodiment has a three-layer structure where the two optical devices **10**, **20** are bonded by the adhesive layer **30**. In contrast, the stacked lens **1**A of Modification 1 of the first embodiment illustrated in FIG. **6** has a seven-layer structure including four optical devices **10**A to **10**D and three adhesive layers **30**A to **30**C.
(55) The optical device **10**A includes a glass device **11**A and a resin lens **12**A. The optical device **10**B includes a glass device **11**B and a resin lens **12**B. The optical device **10**C, which is disposed on an optical axis of the first optical device **10**A, includes a glass device **11**C and a resin lens **12**C. The optical device **10**D, which is disposed on the optical axis of the first optical device **10**A, is a glass device **11**D. Principal surfaces of the optical devices **10**A to **10**D are substantially in the same shape and have substantially the same surface area.
(56) The adhesive layer **30**A bonds the optical device **10**A and the optical device **10**B. The adhesive layer **30**B bonds the optical device **10**B and the optical device **10**C. The adhesive layer **30**C bonds the optical device **10**C and the optical device **10**D.
(57) In other words, the adhesive layer **30**A bonds the resin lens **12**A of the optical device **10**A and the resin lens **12**B of the optical device **10**B. The adhesive layer **30**B bonds the glass device **11**B of the optical device **10**B and the glass device **11**C of the optical device **10**C.
(58) The three adhesive layers **30**A to **30**C include picture-frame-shaped projections **31**A to **31**C projecting from the four side surfaces of the optical devices **10**A to **10**D, respectively.
(59) <Modification 2 of First Embodiment>
(60) The stacked lens **1**B of Modification 2 of the first embodiment illustrated in FIG. **7** includes

the four optical devices **10**A to **10**D, three adhesive layers **30**D to **30**F, and an image pickup device **40** disposed on the optical axis of the first optical device **10**A. The image pickup device **40** is a CCD or the like configured to receive light collected by the four optical devices **10**A to **10**D.

(61) The first adhesive layer **30**D bonds the optical device **10**A and the optical device **10**B. The second adhesive layer **30**E bonds the optical device **10**B and the optical device **10**C. The third adhesive layer **30**F bonds the optical device **10**C and the optical device **10**D. The principal surfaces of the optical devices **10**A to **10**D are substantially in the same shape and have substantially the same surface area.

(62) The first adhesive layer **30**D includes two first projections **31**D projecting in a Y-axis direction. The second adhesive layer **30**E includes two second projections **31**E projecting in an X-axis direction. The second projections **31**E project outside an outer peripheral surface (side surfaces) of each of the second optical device **10**B and the third optical device **10**C along the optical axis O.

(63) In other words, the first projections **31**D and the second projections **31**E of the stacked lens **1**B each include two projecting projections located at symmetric positions across the optical axis O. Note that the third adhesive layer **30**F has no projection from the side surface of the optical member. That is, four side surfaces of the third adhesive layer **30**F are flush with the respective four side surfaces of each of the optical devices **10**A to **10**D.

(64) A plurality of projected images of the two first projections **31**D and the two second projections **31**E that are projected on an imaginary plane orthogonal to the optical axis have no overlap.

(65) The stacked lens **1**B is produced by cutting away a part of projections projecting from the four side surfaces of each of the adhesive layers **30**A, **30**B of the stacked lens **1**A.

(66) In addition, according to a method of manufacturing the stacked lens **1**B, a plurality of image pickup devices **40** are disposed on the stacked wafer **1**W, and when the stacked wafer **1**W is cut, the stacked lens **1**B is cut such that the stacked lens **1**B includes the image pickup device **40**.

(67) The stacked lens **1**B, which includes the image pickup device **40**, has the same effect as the effect of the stacked lens **1**.

(68) In a case where a stacked lens includes three or more optical devices, two or more adhesive layers each bond two of the plurality of optical devices. It is sufficient if at least one of the two or more adhesive layers includes a projection projecting from side surfaces of the plurality of optical devices. In addition, the projection projecting from the side surfaces of the optical devices is not necessarily in the shape of a picture frame surrounding the optical devices.

(69) In other words, a stacked lens of the present invention includes a plurality of optical devices and at least one adhesive layer each bonding two of the plurality of optical devices, at least one of the adhesive layers including a projection projecting outside side surfaces of the plurality of optical devices.

(70) Note that the stacked lens may include a light-emitting device in place of the image pickup device **40**.

Second Embodiment

(71) As illustrated in FIGS. **8** to **10**, an optical unit **2** of the present embodiment includes the stacked lens **1** of the first embodiment, a frame member (a first frame member) **50**, and resin **60** placed to fill a gap between a side surface of the stacked lens **1** and an inner surface of the frame member **50**.

(72) The resin **60** fixes the stacked lens **1** to the frame member **50**. To block outside light entering a light path through the side surface of the stacked lens **1**, it is preferable that the resin **60** be light-blocking resin.

(73) As already described, the stacked lens **1** includes the first optical device **10**, the second optical device **20**, and the adhesive layer **30**. The adhesive layer **30** includes the picture-frame-shaped projection **31** projecting outside the side surfaces of the first optical device **10** and the second optical device **20** with respect to the direction of the optical axis O.

(74) The frame member **50** is a hollow rectangular cylinder having through holes H**50**, H**55**. Cross sections of the through holes H**50**, H**55** orthogonal to the optical axis O have a square shape.
(75) As illustrated in FIG. **8**, as compared with an inner size L**50** of the cross section of the through hole H**50** orthogonal to the optical axis O, an inner size L**55** of the cross section of the through hole H**55** orthogonal to the optical axis O is small. In other words, the through hole H**50** has a third projection **55** (hereinafter, referred to as a "projection **55**") projecting from an inner surface **50**SSA. A wall surface of the projection **55** defines an inner surface of the through hole H**55**.
(76) Cross sections of the first optical device **10** and the second optical device **20** orthogonal to the optical axis O have a square shape. As illustrated in FIG. **8**, an outer size L**30** of the adhesive layer **30**, which includes the picture-frame-shaped projection **31**, is larger than an outer size L**1** of the cross sections of the first optical device **10** and the second optical device **20** orthogonal to the optical axis O.
(77) Further, the outer size L**30** of the adhesive layer **30** is smaller than the inner size L**50** of the through hole **1150** and larger than the inner size L**55** of the through hole H**55**.
(78) Thus, the projection **31** of the adhesive layer **30** of the stacked lens **1** housed in the frame member **50** is in contact with the projection **55** of the frame member **50**. The projection **55** is a positioning member (a first positioning member) configured to determine a relative position between the stacked lens **1** and the frame member **50**. Thus, in the optical unit **2**, the relative position between the stacked lens **1** and the frame member **50** in the optical axis direction is accurately determined. Note that the projection **55** also has an effect to prevent the resin **60** from sticking to the incoming surface of the stacked lens **1**.
(79) Note that for an optical unit including a stacked lens including two or more adhesive layers each including a projection projecting from a side surface of an optical device as the stacked lens **1**A illustrated in FIG. **6**, it is sufficient if one projection of a frame member is in contact with one projection of the adhesive layers.
(80) <Method of Manufacturing Optical Unit>
(81) FIG. **11** illustrates a flowchart of a method of manufacturing the optical unit.
(82) <Steps S**10** to S**30**>
(83) It is the same as the method of manufacturing the stacked lens **1** of the first embodiment as already described.
(84) <Step S**40**> Insertion of Stacked Lens
(85) When each of the plurality of stacked lenses **1**, which are produced by dicing the stacked wafer **1**W, is inserted into the frame member **50** having the inner surface with the projection **55**, the projection **31** of the adhesive layer **30** comes into contact with the projection **55**. That is, the projection **55** is the positioning member configured to determine the relative position between the stacked lens **1** and the frame member **50**.
(86) <Step S**50**> Resin Filling
(87) The light-blocking resin **60** is placed to fill a space between an outer peripheral surface (four side surfaces) of the stacked lens **1** and the inner surface of the frame member **50**. The resin **60** is subjected to a cure treatment while the first principal surface **10**SA is immobilized using, for example, a flat plate so that the first principal surface **10**SA of the stacked lens **1** is flush with a top surface **50**SA of the frame member **50**.
(88) Note that the first principal surface **10**SA of the stacked lens **1** and the top surface **50**SA of the frame member **50** are not necessarily flush with each other. However, in order to more accurately determine the relative position between the stacked lens **1** and the frame member **50**, it is preferable that after positioning using the projection **31** and the projection **55**, the soft projection **31** be pressed to deform to make the first principal surface **10**SA of the stacked lens **1** flush with the top surface **50**SA of the frame member **50** and, in this state, the resin **60** be subjected to the cure treatment.
(89) According to a method of manufacturing the optical unit **2**, the relative position between the

stacked lens **1** and the frame member **50** is accurately determined by virtue of the use of the projection **31** of the adhesive layer **30**, which is easily formed during the manufacturing of the stacked lens **1**. Thus, the optical unit **2** is easy to manufacture. The manufacturing method of the present embodiment enables easy manufacturing and reduced manufacturing costs as compared with a conventional manufacturing method including forming, in a side surface of a stacked lens, a recess for positioning.

(90) <Modifications of Second Embodiment>

(91) Since stacked lenses **2**A, **2**B of Modifications 1, 2 of the second embodiment are similar to the optical unit **2**, the same reference sign is assigned to components with the same function and an explanation is omitted.

(92) <Modification 1 of Second Embodiment>

(93) The optical unit **2**A of the present modification illustrated in FIGS. **12** to **14** includes a stacked lens **1**C, a frame member **50**A, and the resin **60**. Note that the resin **60** is not illustrated in FIG. **12**.

(94) The stacked lens **1**C includes the first optical device **10**, the second optical device **20**, and an adhesive layer **30**G. The first optical device **10** and the second optical device **20** have the same configuration as the configuration of the stacked lens **1** of the first embodiment.

(95) The adhesive layer **30**G of the stacked lens **1**C includes a projection **31**G that is an elongated projection projecting from the side surfaces of the optical devices. The projection **31**G includes the two projections **31**A, **31**B located at the symmetric positions across the optical axis O.

(96) The frame member **50**A includes a frame member **51**A and a frame member **51**B. In the frame member **50**A, two through holes H**51**A, H**51**B are formed from respective two opposite inner surfaces **50**SSA to outer surfaces **50**SSB.

(97) An inner size of the two through holes H**51**A, **1151**B is slightly larger than an outer size of the two projections **31**A, **31**B. The projections **31**A, **31**B are inserted into the through holes H**51**A, H**51**B, respectively.

(98) According to a method of manufacturing the optical unit **2**A, the projection **31**A of the stacked lens **1**C is inserted into the through hole H**51**A of the frame member **51**A. In addition, the projection **31**B of the stacked lens **1**C is inserted into the through hole H**51**B of the frame member **51**B. That is, the frame member **50**A is produced by combining the frame member **51**A and the frame member **51**B such that the stacked lens **1**C is housed.

(99) The through holes H**51**A, H**51**B are positioning members configured to determine a relative position between the stacked lens **1**C and the frame member **50**A.

(100) The optical unit **2**A has the same effect as the optical unit **2**.

(101) Note that the two projections **31**A, **31**B projecting from the outer surfaces **50**SSB of the optical unit **2**A may further be in contact with a second positioning member of another second frame member. The optical unit **2**A is easy to be positioned with respect to the second frame member.

(102) <Modification 2 of Second Embodiment>

(103) The optical unit **2**B of the present modification illustrated in FIG. **15** includes the stacked lens **1**B and a frame member **50**B. As already described, the stacked lens **1**B includes the four optical devices **10**A to **10**D, the three adhesive layers **30**D to **30**F, and the image pickup device **40**.

(104) The first adhesive layer **30**D includes the two first projections **31**D projecting in the Y-axis direction. The second adhesive layer **30**E includes the two second projections **31**E projecting in the X-axis direction.

(105) The frame member **50**B includes projections **55**A, **55**B that are positioning members projecting from the inner surface **50**SSA of the through hole H**50**. The projections **55**A are located at symmetric positions in the Y-axis direction across the optical axis O. The projections **55**B are located at symmetric positions in the X-axis direction across the optical axis O. A plurality of projected images of the two projections **31**D and the two projections **31**E projected on an imaginary plane orthogonal to the optical axis have no overlap.

(106) When the stacked lens **1**B is housed in the through hole H**50** of the frame member **50**B, the first projections **31**D come into contact with the projections **55**A and the second projections **31**E come into contact with the projections **55**B.
(107) Since a relative position between the stacked lens **1**B and the frame member **50**B is determined by the projections **31**D, **31**E at the different positions in the optical axis direction and the projections **55**A, **55**B at the different positions in the optical axis direction, the optical unit **2**B is positioned with a higher accuracy than the optical unit **2**.
Third Embodiment
(108) An endoscope **9** of the present embodiment illustrated in FIG. **16** includes an insertion section **91**, an operation section **92**, a universal cord **93**, and an endoscope connector **94**.
(109) The insertion section **91**, which is in a form of an elongated tube, is to be inserted into a body cavity of a living body. The insertion section **91**, in which a distal end portion **91**A, a bending portion **91**B, and a flexible tube **91**C are disposed continuously in this order from a distal end side, is flexible as a whole.
(110) The distal end portion **91**A includes a rigid member **91**A**1** including various units inside. The various units are an image pickup unit including the optical unit **2** (**2**A, **2**B), a treatment instrument insertion channel, an illumination unit including the optical unit **2** (**2**A, **2**B), and the like.
(111) The bending portion **91**B is bendable in upper, lower, right, and left directions in accordance with a rotary operation of a bending knob of the operation section **92** for performing a bending operation.
(112) The flexible tube **91**C is a passively flexible tubular member with softness. The treatment instrument insertion channel, various electric signal lines, a light guide fiber bundle, and the like are inserted within the flexible tube **91**C. The electric signal lines are drawn from the image pickup unit installed in the distal end portion **91**A, being routed to the universal cord **93** through the operation section **92**. The light guide fiber bundle is configured to guide light from a light source apparatus, which is an external apparatus, to a distal end surface of the distal end portion **91**A.
(113) The operation section **92**, which is disposed continuously to a proximal end portion of the insertion section **91**, includes a plurality of operation members and the like. The universal cord **93** is a tubular member having flexibility and that is drawn from the operation section **92**. The endoscope connector **94** is a connection member for connecting the universal cord **93** and the external apparatus.
(114) The endoscope **9** includes the optical unit **2** (**2**A, **2**B) disposed in the distal end portion **91**A of the insertion section **91**. As already described, the relative position between the stacked lens and the frame member is accurately determined, so that the optical unit **2** (**2**A, **2**B) is excellent in optical properties.
(115) Note that the optical unit may be housed in a hole of the rigid member **91**A**1** of the distal end portion **91**A or the rigid member **91**A**1** may be used as a frame member where the stacked lens is housed.
(116) For example, in an endoscope illustrated in FIG. **17**, the rigid member **91**A**1** (the frame member) and a stacked lens **1**D constitute an optical unit **2**C. The rigid member **91**A**1** has a first opening H**91**A in which the stacked lens **1**D is to be housed and a channel opening H**91**B. An inwardly projecting projection (a positioning member) **55**C is formed in an inner surface of the first opening H**91**A. Note that the projection **55**C is in a form of an elongated bank. A hole for housing an illumination optical system and the like, which is not illustrated, is also formed in the rigid member **91**A**1**. The adhesive layer **30**G of the stacked lens **1**D includes a projection **31**G projecting from the side surfaces of the optical devices.
(117) When the stacked lens **1**D is inserted into the first opening H**91**A, the projection **31**G of the adhesive layer **30**G comes into contact with the projection (the positioning member) **55**C. Thus, a relative position between the stacked lens **1**D and the distal end portion **91**A (the frame member) in the optical axis direction is accurately determined. Thus, the optical unit **2**C is excellent in optical

properties.

(118) In addition, since machining for forming an engagement portion, such as a recessed portion, in the stacked lens after blade dicing machining is not necessary, the optical unit **2**C is easy to manufacture and a reduction in manufacturing costs can be expected. In addition, since the stacked lens **1**D is directly attached to the rigid member **91**A**1**, the distal end portion **91**A of the endoscope has a small diameter.

(119) Note that the endoscope of the embodiment may be a flexible endoscope including a flexible insertion section or a rigid endoscope including a rigid insertion section. In addition, the endoscope of the embodiment may be used for medical purposes or industrial purposes.

(120) The present invention is not limited to the above-described embodiments and various modifications, improvements, and the like are possible unless the gist of the present invention is changed.

## Claims

1. A stacked lens comprising: a first optical device; a second optical device disposed on an optical axis of the first optical device; and a first adhesive layer bonding the first optical device and the second optical device and including a first projection projecting outside an outer peripheral surface of each of the first optical device and the second optical device, the outer peripheral surface being along the optical axis.
2. The stacked lens according to claim 1, wherein the outer peripheral surface includes four side surfaces of each of the first optical device and the second optical device, and the first projection projects from at least one of the four side surfaces.
3. The stacked lens according to claim 2, wherein the first projection projects from the four side surfaces.
4. The stacked lens according to claim 1, wherein the outer peripheral surface of at least one of the first optical device or the second optical device is a cut surface including a striation.
5. The stacked lens according to claim 1, wherein at least one of the first optical device or the second optical device is a hybrid lens device including a glass and a resin lens.
6. The stacked lens according to claim 1, wherein the adhesive layer is made of transparent resin.
7. The stacked lens according to claim 1, comprising an image pickup device configured to receive light collected by the first optical device and the second optical device.
8. The stacked lens according to claim 1, further comprising a third optical device and a second adhesive layer bonding the second optical device and the third optical device, wherein the second adhesive layer includes a second projection, the second projection projecting outside the outer peripheral surface of the second optical device, the outer peripheral surface being along the optical axis, the second projection projecting outside an outer peripheral surface of the third optical device, the outer peripheral surface being along the optical axis, and a projected image of the first projection and a projected image of the second projection that are projected on an imaginary plane orthogonal to the optical axis have no overlap.
9. An optical unit comprising: a stacked lens including: a first optical device; a second optical device disposed on an optical axis of the first optical device; and a first adhesive layer bonding the first optical device and the second optical device and including a first projection projecting outside an outer peripheral surface of each of the first optical device and the second optical device, the outer peripheral surface being along the optical axis; and a first frame member in which the stacked lens is housed and that has an inner surface including a first positioning member, the first projection being in contact with the first positioning member.
10. The optical unit according to claim 9, wherein the first positioning member includes a third projection projecting from the inner surface.
11. The optical unit according to claim 9, wherein the first positioning member includes a through

hole from the inner surface to an outer surface of the first frame member and through which the first projection penetrates.

12. The optical unit according to claim 9, further comprising a second frame member including a second positioning member being in contact with the first projection projecting from an outer surface of the first frame member.

13. The optical unit according to claim 9, wherein the stacked lens further includes a third optical device disposed on the optical axis of the first optical device and a second adhesive layer bonding the second optical device and the third optical device, the second adhesive layer includes a second projection, the second projection projecting outside the outer peripheral surface of the second optical device, the outer peripheral surface being along the optical axis, the second projection projecting outside an outer peripheral surface of the third optical device, the outer peripheral surface being along the optical axis, a projected image of the first projection and a projected image of the second projection that are projected on an imaginary plane orthogonal to the optical axis have no overlap, and the first frame member includes a plurality of positioning members, the plurality of positioning members each being in contact with the first projection or the second projection.

14. An endoscope comprising an insertion section and an optical unit disposed in a distal end portion of the insertion section, the optical unit including: a stacked lens including: a first optical device; a second optical device disposed on an optical axis of the first optical device; and a first adhesive layer bonding the first optical device and the second optical device and including a first projection projecting outside an outer peripheral surface of each of the first optical device and the second optical device, the outer peripheral surface being along the optical axis; and a frame member in which the stacked lens is housed and that has an inner surface including a positioning member, the first projection being in contact with the positioning member.

15. A method of manufacturing an optical unit, the method comprising: producing a stacked wafer including a first optical wafer including a plurality of first optical devices, a second optical wafer including a plurality of second optical devices, and an adhesive layer between the first optical wafer and the second optical wafer; cutting the stacked wafer using a dicing blade to produce a stacked lens, the stacked lens including one of the first optical devices, one of the second optical devices, and a first adhesive layer including a projection projecting outside a cut surface; and housing the stacked lens in a frame member having an inner surface including a positioning member in a state where the projection is in contact with the positioning member.

16. The method of manufacturing an optical unit according to claim 15, the method comprising placing resin to fill a space between a side surface of the stacked lens and the inner surface of the frame member.

17. The method of manufacturing an optical unit according to claim 15, the method comprising: disposing an image pickup device in the stacked wafer before the cutting of the stacked wafer; and cutting the stacked wafer in a state where the stacked lens includes the image pickup device.